



Workshop Announcement: Technical Updates on Power and Energy Systems April 20, 2018

Co-sponsored by: The Electric Energy Risk Research Laboratory,
The Laboratory for Electrical Energy & Power Systems,
and the School of Electrical and Computer Engineering, at The University of Oklahoma

Synopsis: The rate of technological advancement in power systems has been on an ever increasing trajectory over the past few years. There has been significant interest and growth in distributed energy resources, intelligent grid technologies and microgrids, particularly those operating with combined heat and power systems such as buildings and districts. These developments come at a time of rising threats in cybersecurity and disturbances as has been seen in the State of Oklahoma and worldwide. Innovation is needed on several fronts such as developing flexible and resilient networks integrated with advanced control, automation and protection systems.

This one-day technical workshop is intended to provide an update on several frontier issues impacting the development of distributed energy resources and microgrids from a grid resiliency and security perspective. Five speakers will provide technical updates on various technologies related to microgrids, distributed energy resources, energy management, protection and control. People who are interested in these emerging issues and technologies, including engineers, regulators and policy makes, will benefit from the technical updates and discussions at the workshop.

* Certificate of 0.5 Continuing Education Credits (CEU) will be issued by University OUTREACH, OU College of Continuing Education

Topics/Speakers

- Transformer protection in modern transmission and distribution networks – **By Jared Candelaria**, Protection Application Engineer, Schweitzer Engineering Laboratories, Inc.
- Phase-shifting transformer for managing congestion issues in Woodward, Oklahoma – **By Travis Hyde**, Director of Transmission and Distribution, Oklahoma Gas & Electric
- A FPGA-based hardware-in-the-loop (HIL) simulation and control platform for microgrids and smart grids with DERs – **By Igor Alvarado**, Business Development Manager, National Instruments
- Energy management and control system solutions for microgrids; functionality, value, benefits and lessons learned – **By Maggie Clout**, Microgrid Business Development Manager, Siemens
- Using Buildings as Virtual Energy Storage: Demand Response and Ancillary Services – **By Jie Cai**, University of Oklahoma and Johnson Controls

Workshop Organizers - Executive Committee

John N. Jiang (University of Oklahoma, OG&E Professor)

Paul Moses (University of Oklahoma, Assistant Professor)

Travis Hyde (OG&E Corp, Committee Chair)

Leon Howell (OG&E Corp, Committee Member)

JR Cruz (University of Oklahoma, ECE Dept. Director and Chair)

Tentative Workshop Agenda

Date: Friday, April 20, 2018

Location: Devon Energy Hall 110 W. Boyd Street,
Room 420, Norman, Oklahoma 73019

9:20am

Sign-in (light refreshment and beverages served)

9:30am

Welcome remarks from *JR Cruz*, Director, School of Electrical and Computer Engineering, The University of Oklahoma

9:45-10:45am

Transformer protection in modern transmission and distribution networks – By Jared Candelaria, Schweitzer Engineering Laboratories, Inc.

10:45-11:45am

Phase-shifting transformer for managing congestion issues in Woodward, Oklahoma – By Travis Hyde, Oklahoma Gas & Electric

11:45-12:45pm Lunch Break (provided)

12:45-1:45pm

A FPGA-based hardware-in-the-loop (HIL) simulation and control platform for microgrids and smart grids with DERs – By Igor Alvarado, National Instruments

1:45-2:45pm

Energy management and control system solutions for microgrids; functionality, value, benefits and lessons learned – By Maggie Clout - Siemens

2:45-3:00pm Coffee Break

3:00-4:00pm

Using Buildings as Virtual Energy Storage: Demand Response and Ancillary Services– By Jie Cai, University of Oklahoma and Johnson Controls

4:15 pm Adjourn

Registration Information:

Participants may register online (the registration page link will be posted in the near future).

Or onsite, the day of the workshop. Registration is \$75 per person and \$60 per person in a group more than 5. There is no cancellation refund for the workshop.

Parking:

All parking on the OU campus is by permit only. Please download and print the permit from the registration page, to allow you to park in faculty/staff lot for day of the workshop; or you can pick one up in the meeting room on the morning of the workshop. The closest parking for Devon Energy Hall is south of the Sarkey's Energy Center. A link to a map will be available at the registration site.

For information on Registration and Event Logistics, please contact Cathy Trujillo at 405-325-4723 or cathyt@ou.edu

